



Class: TY B.Tech.

Course: Metal Cutting Theory

Course Code: PCC-PE305

Date: 18.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Questions	Marks	CO	BT
Q.1	A Draw single point cutting tool and its different views with all nomenclature.	4	CO3	BT2
	B Write a detailed note about High-Speed Steel (HSS) tool material.	4	CO1	BT1
	C How various machining parameters affect tool life, machinability index?	4	CO2	BT3
Q.2	A In an orthogonal cutting process, the following observations were made: Depth of cut = 0.25 mm Chip thickness ratio = 0.45 Width of cut = 4 mm Cutting velocity = 40 m/min Cutting force component parallel to cutting velocity = 1150 N Cutting force component normal to cutting velocity = 140 N Rake angle = 18°. Determine the resultant cutting force, power of cutting, shear lane angle, friction angle and force component parallel to shear plane.	4	CO5	BT3
	B What is machinability? How is it measure and how machining parameters affect machinability? A "Through-away" carbide insert was used to machine a steel workpiece with a cutting speed of 60 m/min and a tool life of 40 min. When the speed was increased to 100m/min, the tool life was reduced to 10 min. Using this data, estimate the cutting speed for maximum productivity, if the tool change time is 2 min.	4	CO2	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	The students shall have the knowledge of fundamentals of metal cutting technology including the process, measurements, design and selection of various cutting tools and their industrial specifications.
CO 2	Select suitable cutting tool for performing identified machining operation.
CO 3	Design form tool for given application and conditions.
CO 4	Design single and multiple point cutting tools for given conditions and applications.
CO 5	Understand cutting forces generated during different cutting operations.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.	Course: Machine Design	Course Code: PCC-PE302
Date: 19.08.2025	Time: 15:00 to 15:45	Max. Marks: 20

Note:

- i. *Attempt all the questions.*
- ii. *Attempt any One from Q.1.*
- iii. *Assume suitable data whenever necessary and state it clearly.*
- iv. *Mobile phones and programmable calculators are strictly prohibited.*
- v. *Exchange/Sharing of stationery, calculator etc. not allowed.*
- vi. *Figures to right indicate full marks.*
- vii. *The symbols carry usual meaning.*
- viii. *Use of non-programmable calculator is allowed.*

Q.N.		Marks	CO	BT
	Solve the following questions.			
Q.1	Explain the phases and iterations of the design process in detail with the help of flowchart.	10	CO1	BT1 ,2
	OR			
Q.1	Give detailed explanation about general considerations in machine design with relevant examples.	10	CO1	BT1 ,3
Q.2	A manufacturer is interested in starting a business with five different models of cars ranging from 8.5 to 85 kW capacities. Specify power capacities of the models. There is an expansion plan to further increase the number of models from five to nine to fulfil the requirement of customers. Specify the power capacities of the additional models.	10	CO1	BT4 ,5

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Understand basic design procedure and the necessary design considerations.
CO2	Define and distinguish static and fluctuating loads/stresses.
CO3	Derive empirical relations for different machine elements using relevant theories of failure.
CO4	Analyse behaviour of machine components under static and fluctuating loads/stresses.
CO5	Design shafts, gears, clutches, power screws etc. for real life applications when subjected to static loading using generative design.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engg.
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Metallurgy and Metal
Forming.

Course Code: PCC-PE301

Date: 20.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

Reg.No.....

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- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Write any four differences between ferrous and non-ferrous metal?	02	CO1	BT1
	b	Enlist iron ores? What is pig iron?	02	CO1	BT1
	c	Define the following terms: Creep, Toughness, Plasticity and Strength.	04	CO2	BT2
	d	What are the advantages and disadvantages of Ceramics.	02	CO2	BT1
Q.2		Draw Fe-C equilibrium diagram and label the temperature, composition and phases, also explain Peritectic, Eutectic and Eutectoid transformation reaction.	10	CO1	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Classify the cast irons and their applications and heat treatment processes.
CO2	Describe powder metallurgy, its processes and applications.
CO3	Select the suitable non-destructive testing method for inspecting components
CO4	Classify different metal forming processes.
CO5	Design dies for cutting, forging, bending, drawing, piercing, and blanking.